

ASYN4 Machine Performance Report

Field	Value
NAME	CAGE MOTOR
REKE	SC317
DATUM	17.09.26

Partial-Load Performance

Load	P_mech (kW)	Eta	cos(phi)	I1 (A)	Slip (%)	Pcu1 (kW)	Pcu2 (kW)	M (Nm)
1/4	90.08	0.8956	0.827	168.24		5.443	3.479	1486.4
2/4	22.47	0.9263	0.758	44.44		0.380	0.159	360.0
3/4	45.05	0.9357	0.866	77.13		1.144	0.654	726.8
4/4 (rated)	67.72	0.9230	0.870	116.91		2.628	1.622	1102.3
5/4	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

Breakdown / Locked-Rotor Summary

Quantity	Value
Breakdown slip (Kipp)	6.886 %
Breakdown torque	1759.4 Nm
Locked-rotor torque (theoretical)	775.1 Nm
Locked-rotor current (theoretical)	398.7 A
Locked-rotor torque (corrected, KKA=0.50)	1072.9 Nm
Locked-rotor current (corrected)	472.2 A
Time constant TK	0.0414 s
Time constant THAAPT (rotor)	0.6479 s
Time constant TSTREU (leakage)	0.0110 s

Cooling & Material Information

Quantity	Value
Dominant cooling path (IKLINK)	0
Stator yoke iron area AR1	0.03794 m2
Stator tooth iron area AZ1	0.02442 m2
Rotor yoke iron area AR2	0.07255 m2
Rotor tooth iron area AZ2	0.02963 m2